# Solving Tensor Linear Systems under the t-product by a Kaczmarz-type method

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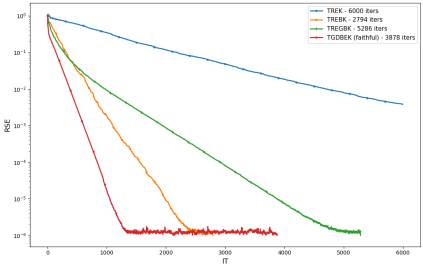
# Why Tensors?

- Images, videos, and multi-dimensional data are naturally represented as **third-order tensors** $\mathcal{A} \in \mathbb{R}^{m \times n \times p}$.
- **Image reconstruction**: given a blurred/noisy observation $\mathcal{B}$, recover the clean image $\mathcal{X}$ from

$$\mathcal{A} * \mathcal{X} = \mathcal{B}$$

  where $*$ denotes the *t-product*.
- Systems are often **large-scale** and **inconsistent**.




Convergence on `Cities` matrix

# Problem Setup

## Tensor Linear System

Given $\mathcal{A} \in \mathbb{R}^{m \times n \times p}$ and $\mathcal{B} \in \mathbb{R}^{m \times k \times p}$, find $\mathcal{X} \in \mathbb{R}^{n \times k \times p}$ satisfying:

$$\mathcal{A} * \mathcal{X} = \mathcal{B}$$

**Goal:** compute the *minimum-norm least-squares* solution

$$\mathcal{X}^\star = \underset{\mathcal{X}}{\operatorname{argmin}} \|\mathcal{X}\|_F \quad \text{s.t.} \quad \mathcal{X} \in \underset{\mathcal{Z}}{\operatorname{argmin}} \|\mathcal{A} * \mathcal{Z} - \mathcal{B}\|_F$$

**Setting:**

- System can be **inconsistent** ($\mathcal{B} \notin \operatorname{range}(\mathcal{A})$)
- Dimensions $m, n$ large $\Rightarrow$ direct solvers impractical
- Iterative Kaczmarz-type methods are well-suited

# Existing Methods & Limitations

**Tensor Kaczmarz variants in the literature:**

- **TREK** – Tensor Randomized Extended Kaczmarz
- **TREBK** – Tensor Randomized Extended Block Kaczmarz
- **TREGBK** – Tensor Randomized Extended Greedy Block Kaczmarz

**Key limitation:**

Pre-defined Partitions Required

TREBK, TREGBK, and similar block methods require a **pre-defined partition** of the row/column indices.

- Partition quality directly affects convergence
- No principled way to choose the partition a priori
- Extra tuning overhead

# TGDBEK: Key Idea

**Two-step greedy selection** at each iteration $k$:

**Z-step** (column selection):

- Score each column slice by its contribution
- Greedily select the set $U_k$ of most informative columns
- Project: $\mathcal{Z}^{k+1} \leftarrow \mathcal{Z}^k - \mathcal{A}_{:,U_k,:} * (\mathcal{A}_{:,U_k,:})^\dagger * \mathcal{Z}^k$

**X-step** (row selection):

- Score each row slice by its residual
- Greedily select the set $J_k$ of most informative rows
- Update: $\mathcal{X}^{k+1} \leftarrow \mathcal{X}^k + (\mathcal{A}_{J_k,:,:})^\dagger * \mathcal{R}_{J_k}$

**No pre-defined partitions needed!**

**Algorithm 1:** Tensor Greedy Double Block Extended Kaczmarz (TGDBEK)

**Input**: $\mathcal{A} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$, $\mathcal{B} \in \mathbb{R}^{N_1 \times K \times N_3}$, iterations $\ell$, parameter $\eta \in (0, 1]$.

**Initialize**: $\mathcal{X}^{(0)} = \mathcal{O}$ and $\mathcal{Z}^{(0)} = \mathcal{B}$.

**for** $k = 0, 1, \ldots, \ell - 1$ **do**

$$\varepsilon_k^z = \eta \max_{1 \le j \le N_2} \left\{ \frac{\left\| (\mathcal{A}_{:,j,:})^\top * \mathcal{Z}^{(k)} \right\|_F^2}{\left\| \mathcal{A}_{:,j,:} \right\|_F^2} \right\}$$

$$U_k = \left\{ j \in [N_2] : \left\| (\mathcal{A}_{:,j,:})^\top * \mathcal{Z}^{(k)} \right\|_F^2 \ge \varepsilon_k^z \left\| \mathcal{A}_{:,j,:} \right\|_F^2 \right\}$$

Update $\mathcal{Z}^{(k+1)} = \mathcal{Z}^{(k)} - \mathcal{A}_{:,U_k,:} * (\mathcal{A}_{:,U_k,:})^\dagger * \mathcal{Z}^{(k)}$

$$\varepsilon_k^x = \eta \max_{1 \le i \le N_1} \left\{ \frac{\left\| \mathcal{B}_{i,:,:} - \mathcal{Z}_{i,:,:}^{(k+1)} - \mathcal{A}_{i,:,:} * \mathcal{X}^{(k)} \right\|_F^2}{\left\| \mathcal{A}_{i,:,:} \right\|_F^2} \right\}$$

$$J_k = \left\{ i \in [N_1] : \left\| \mathcal{B}_{i,:,:} - \mathcal{Z}_{i,:,:}^{(k+1)} - \mathcal{A}_{i,:,:} * \mathcal{X}^{(k)} \right\|_F^2 \ge \varepsilon_k^x \left\| \mathcal{A}_{i,:,:} \right\|_F^2 \right\}$$

Update $\mathcal{X}^{(k+1)} = \mathcal{X}^{(k)} + (\mathcal{A}(J_k,:,:))^\dagger * \left( \mathcal{B}_{J_k,:,:} - \mathcal{Z}_{J_k,:,:}^{(k+1)} - \mathcal{A}(J_k,:,:) * \mathcal{X}^{(k)} \right)$

**end**

The TGDBEK Algorithm

# Greedy Selection Criterion

Given a parameter $\eta \in (0, 1]$:

**Column selection (Z-step):**

$$\varepsilon_z^k = \eta \cdot \max_j \frac{\|(\mathcal{A}_{:,j,:})^\top * \mathcal{Z}^k\|_F^2}{\|\mathcal{A}_{:,j,:}\|_F^2}, \quad U_k = \left\{ j : \frac{\|(\mathcal{A}_{:,j,:})^\top * \mathcal{Z}^k\|_F^2}{\|\mathcal{A}_{:,j,:}\|_F^2} \geq \varepsilon_z^k \right\}$$

**Row selection (X-step):**

$$\varepsilon_x^k = \eta \cdot \max_i \frac{\|\mathcal{B}_{i,:,:} - \mathcal{Z}_{i,:,:}^{k+1} - \mathcal{A}_{i,:,:} * \mathcal{X}'^k\|_F^2}{\|\mathcal{A}_{i,:,:}\|_F^2}, \quad J_k = \left\{ i : \text{score}_i \geq \varepsilon_x^k \right\}$$

- $\eta \to 1$: fewer, most informative slices (aggressive)
- $\eta \to 0$: more slices selected (conservative)
- $U_k, J_k$ are **determined adaptively** at each iteration

# Convergence Guarantee

### Theorem

*Let $\mathcal{A} \in \mathbb{R}^{m \times n \times p}$ and $\mathcal{B} \in \mathbb{R}^{m \times k \times p}$. The iterates $\{\mathcal{X}^k\}$ produced by TGDBEK converge* ***linearly*** *to the minimum-norm least-squares solution $\mathcal{X}^\star$:*

$$\mathbb{E}[\|\mathcal{X}^k - \mathcal{X}^\star\|_F^2] \;\leq\; \rho^k \, \|\mathcal{X}^0 - \mathcal{X}^\star\|_F^2$$

*where $0 < \rho < 1$ depends on $\eta$ and the spectral properties of $\mathcal{A}$.*

**Key points:**

- Convergence to the *minimum-norm least-squares* solution
- Works for inconsistent systems
- Rate improves with well-conditioned $\mathcal{A}$ and appropriate $\eta$

*Full proof available in the technical report.*

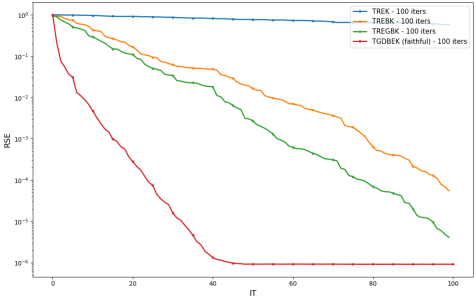
# Results on Sparse Matrices (SuiteSparse)

**ash85** ($85 \times 85$, density $= 0.072$)

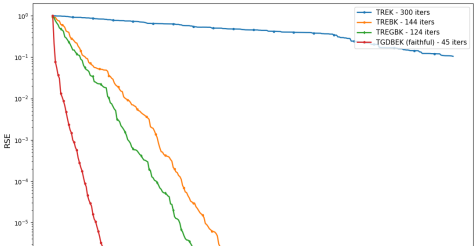
| **Method** | **Time** | **RSE** | **IT** |
|---|---|---|---|
| TREK | 0.370 | 2.0e−2 | 500 |
| TREBK | 0.369 | 9.2e−7 | 174 |
| TREGBK | 0.574 | 9.9e−7 | 258 |
| **TGDBEK** | **0.248** | 9.3e−7 | **78** |

**nos5** ($153 \times 153$, density $= 0.047$)

| **Method** | **Time** | **RSE** | **IT** |
|---|---|---|---|
| TREK | 0.454 | 1.0e−1 | 300 |
| TREBK | 0.860 | 9.8e−7 | 144 |
| TREGBK | 0.864 | 9.9e−7 | 124 |
| **TGDBEK** | **0.581** | 9.7e−7 | **45** |

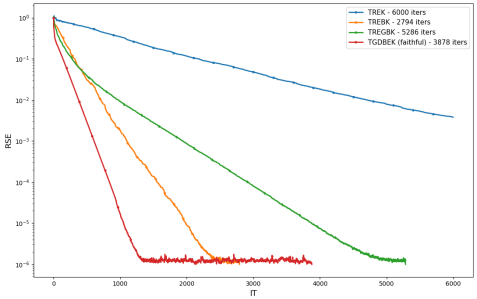



RSE vs. Iterations (ash85)



# Results: Cities Matrix ($128 \times 128$)

| Method | Time | RSE | IT |
|---|---|---|---|
| TREK | 4.089 | $3.8\mathrm{e}{-3}$ | 6000 |
| TREBK | 3.318 | $9.9\mathrm{e}{-7}$ | 2794 |
| TREGBK | 6.943 | $9.9\mathrm{e}{-7}$ | 5286 |
| **TGDBEK** | 5.954 | $9.9\mathrm{e}{-7}$ | **3878** |




RSE vs. Iterations (Cities)

**Takeaway:**

- TGDBEK consistently reaches target accuracy with **fewer iterations**
- Competitive or better wall-clock time **without needing partition tuning**

# Summary & Future Work

**Summary:**

1. **No pre-defined partitions** – greedy selection adapts at each step
2. **Faster convergence** – fewer iterations than TREK, TREBK, TREGBK
3. **Proven guarantees** – linear convergence to min-norm LS solution

**Future work:**

- Larger-scale experiments (high-resolution images, video)
- GPU-accelerated implementation via PyTorch
- Extensions to other tensor decompositions (Tucker, TT)
- Applications to tensor completion and compressed sensing

**Thank you!**

Code:

`https://github.com/jnlandu/tensor-greedy-double-extended-kaczmarz`